

1 Categories

• A *category* consists of a collection of *objects* and of *morphisms* between these objects satisfying some conditions.

Definition 1. A **category** $\mathbf{C}$ consists of

- (1) a class of **objects** of the category, which is denoted by $\text{Obj}(\mathbf{C})$;
- (2) for any two objects A, B of the category $\mathbf{C}$, there is the set $\text{Hom}_{\mathbf{C}}(A, B)$, whose elements are called **morphisms**, with the properties below:

a) for any object A in $\mathbf{C}$, there is at least one morphism $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$, called the **identity morphism**;

b) for any two morphisms $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}}(B, C)$, they determine a morphism $g \circ f \in \text{Hom}_{\mathbf{C}}(A, C)$. i.e., for any triple of objects A, B, C of $\mathbf{C}$, there is the function

$$\text{Hom}_{\mathbf{C}}(A, B) \times \text{Hom}_{\mathbf{C}}(B, C) \longrightarrow \text{Hom}_{\mathbf{C}}(A, C)$$

$$(f, g) \mapsto g \circ f;$$

c) for any $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$ and $h \in \text{Hom}_{\mathbf{C}}(C, D)$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f \in \text{Hom}_{\mathbf{C}}(A, D),$$

i.e., the composition law is associative;

d) for any $f \in \text{Hom}_{\mathbf{C}}(A, B)$, we have

$$f \circ 1_A = f \quad \text{and} \quad 1_B \circ f = f;$$

e) the sets $\text{Hom}_{\mathbf{C}}(A, B) \cap \text{Hom}_{\mathbf{C}}(C, D) = \emptyset$ unless $A = C, B = D$.

Remark: (A) To remember all these properties above, we can just think of the properties of maps of sets;

(B) A morphism $f \in \text{Hom}_{\mathbf{C}}(A, A)$ will be called an **endomorphism** and $\text{End}_{\mathbf{C}}(A) := \text{Hom}_{\mathbf{C}}(A, A)$; If the category is clear in the text, we will simply drop the index $\mathbf{C}$ in the notation $\text{Hom}_{\mathbf{C}}(A, B)$; For a morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$, we may simply write

$$f : A \longrightarrow B \quad \text{or} \quad A \xrightarrow{f} B.$$

Example 1. Let **Set** be the category such that

$$\text{Obj}(\mathbf{Set}) := \text{the class of all sets},$$

$$\text{Hom}_{\mathbf{Set}}(A, B) := B^A := \{f \mid f : A \longrightarrow B \text{ is a map between the two sets}\},$$

for any two A, B in $\text{Obj}(\mathbf{Set})$.

Example 2. Assume S is a set and let $\sim$ be a relation on S such that it is reflexive and transitive, we define a category $\mathbf{S}$ as below:

$$\text{Obj}(\mathbf{S}) := \text{elements of } S,$$

$$\text{Hom}_{\mathbf{S}}(s_1, s_2) := \begin{cases} \{(s_1, s_2)\} (\subseteq S \times S), & \text{if } s_1 \sim s_2 \\ \emptyset, & \text{otherwise.} \end{cases}$$

For any element $s \in S$, $1_s := (s, s) \in \text{Hom}_{\mathbf{S}}(s, s)$. For $s_1, s_2, s_3 \in S$, if $s_1 \sim s_2$ and $s_2 \sim s_3$, then $s_1 \sim s_3$ by the transitivity, then for $f = (s_1, s_2) \in \text{Hom}_{\mathbf{S}}(s_1, s_2)$, $g = (s_2, s_3) \in \text{Hom}_{\mathbf{S}}(s_2, s_3)$, $g \circ f := (s_1, s_3) \in \text{Hom}_{\mathbf{S}}(s_1, s_3)$. We can check that these indeed make $\mathbf{S}$ a category.

Example 3. Let $\mathbf{C}$ be a category and let A be an object in $\text{Obj}(\mathbf{C})$. We define a new category $\mathbf{C}_A$ as below:

$$\text{Obj}(\mathbf{C}_A) := \text{all morphisms from some object of } \mathbf{C} \text{ to } A,$$

i.e., an object in $\text{Obj}(\mathbf{C}_A)$ is a morphism $f \in \text{Hom}_{\mathbf{C}}(B, A)$ for some object B in $\text{Obj}(\mathbf{C})$.

Pictorially, we may write

$$\begin{array}{c} B \\ \downarrow f \\ A \end{array}$$

Now, for two objects $f_1 \in \text{Hom}_{\mathbf{C}}(B_1, A)$ and $f_2 \in \text{Hom}_{\mathbf{C}}(B_2, A)$ of $\mathbf{C}_A$

$$\begin{array}{ccc} B_1 & & B_2 \\ f_1 \downarrow & & f_2 \downarrow \\ A & & A \end{array}$$

we define a morphism $\sigma \in \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$ to be a morphism $\sigma \in \text{Hom}_{\mathbf{C}}(B_1, B_2)$ in the category $\mathbf{C}$ such that the following diagram is commutative

$$\begin{array}{ccc} B_1 & \xrightarrow{\sigma} & B_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

i.e., $f_2 \circ \sigma = f_1$. Hence for two morphisms $\sigma \in \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$, $\tau \in \text{Hom}_{\mathbf{C}_A}(f_2, f_3)$, the composition morphism $\tau \circ \sigma \in \text{Hom}_{\mathbf{C}_A}(f_1, f_3)$ is the one that makes the following diagram commutative

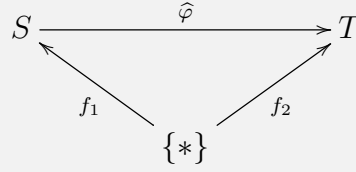
$$\begin{array}{ccccc} B_1 & \xrightarrow{\sigma} & B_2 & \xrightarrow{\tau} & B_3 \\ & \searrow f_1 & \downarrow f_2 & \swarrow f_3 & \\ & & A & & \end{array}$$

$\tau \circ \sigma$ (curved arrow from B_1 to B_3)

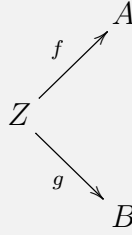
then we see that such defined $\mathbf{C}_A$ is a category. Categories constructed in such way are called **slice categories**.

Example 4. A similar example to the example above can be obtained by considering morphisms in a category $\mathcal{C}$ from a fixed object A to all other objects in $\mathcal{C}$, again with morphisms defined by suitable commutative diagrams. Such categories are called **coslice categories**, denoted as $\mathcal{C}^A$.

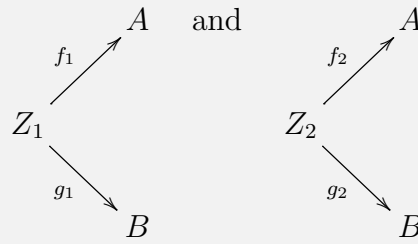
Example 5. We give a concrete instance of a category as in Example 4, we define $\mathcal{C} := \mathbf{Set}$ and let A be a fixed singleton $A = \{*\}$, then resulting category is denoted by $\mathbf{Set}^* := \mathbf{Set}^{\{*\}}$. Hence, an object in $\mathbf{Obj}(\mathbf{Set}^*)$ is a morphism $f \in \mathbf{Hom}_{\mathbf{Set}}(\{*\}, S)$ for some set S . Thus, we may denote an object of $\mathbf{Set}^*$ as (S, s) , where $s = f(*) \in S$. Then we immediately see that, a morphism $\varphi \in \mathbf{Hom}_{\mathbf{Set}^*}((S, s), (T, t))$ corresponds to a map $\hat{\varphi} : S \rightarrow T$ such that $\hat{\varphi}(s) = t$.



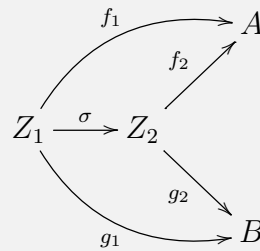
Example 6. Let $\mathcal{C}$ be a category and let A, B be two given objects in $\mathbf{Obj}(\mathcal{C})$. Now we define a new category $\mathcal{C}_{A,B}$ as below: define $\mathbf{Obj}(\mathcal{C}_{A,B}) :=$ diagrams



and define a morphism σ between

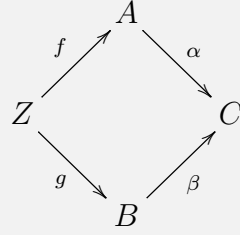


to be the commutative diagram

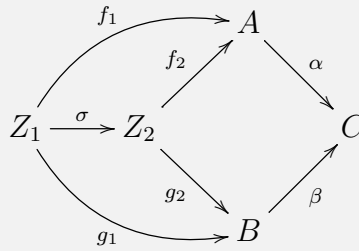


i.e., $f_2 \circ \sigma = f_1$ and $g_2 \circ \sigma = g_1$. So the category $\mathcal{C}_{A,B}$ is actually a mixture of $\mathcal{C}_A$ and $\mathcal{C}_B$. Similarly, we can define $\mathcal{C}^{A,B}$ by flipping the arrows.

Example 7. Let $\mathbf{C}$ be a category and take two fixed morphisms $\alpha : A \rightarrow C, \beta : B \rightarrow C$ in $\mathbf{C}$, where A, B, C are objects in $\text{Obj}(\mathbf{C})$. We define a category $\mathbf{C}_{\alpha, \beta}$ as follows: define $\text{Obj}(\mathbf{C}_{\alpha, \beta}) :=$ commutative diagrams



in $\mathbf{C}$ and define morphisms which actually corresponds to commutative diagrams



Similarly, we can define a category $\mathbf{C}^{\alpha, \beta}$.

2 Morphisms

Definition 2. Let $\mathbf{C}$ be a category, a morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is called an **isomorphism** if it has two-sided inverse, which means that $\exists g \in \text{Hom}_{\mathbf{C}}(B, A)$ such that

$$g \circ f = 1_A \text{ and } f \circ g = 1_B.$$

Remark: If $f \in \text{Hom}_{\mathbf{C}}(A, A)$ is an isomorphism, then we call f an **automorphism** from A to itself. The set of all automorphisms of A is denoted as $\text{Aut}_{\mathbf{C}}(A)$, i.e.,

$$\text{Aut}_{\mathbf{C}}(A) := \{f \in \text{Hom}_{\mathbf{C}}(A, A) \mid f \text{ is an isomorphism}\},$$

which is a subset of $\text{End}_{\mathbf{C}}(A)$.

Proposition 1. The inverse of an isomorphism is unique.

Proof: Let $f \in \text{Hom}_{\mathbf{C}}(A, B)$ be an isomorphism with inverse $g_1, g_2 \in \text{Hom}_{\mathbf{C}}(B, A)$, then we see that

$$g_1 = g_1 \circ 1_B = g_1 \circ (f \circ g_2) = (g_1 \circ f) \circ g_2 = 1_A \circ g_2 = g_2,$$

which completes the proof. ■

Remark: In the proof, actually we see that if a morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ has a left-inverse g_1 and a right-inverse g_2 , then $g_1 = g_2$ and this morphism must be the unique inverse of f . The inverse of an isomorphism f will be denoted as f^{-1} .

The following list of peoperties of the inverse of isomorphisms is not hard to prove.

Proposition 2. Let $\mathcal{C}$ be a category,

- (1) each identity morphism 1_A is an isomorphism and $1_A^{-1} = 1_A$;
- (2) if $f \in \text{Hom}_{\mathcal{C}}(A, B)$ is an isomoprhim, then $f^{-1} \in \text{Hom}_{\mathcal{C}}(B, A)$ is also an isomorphism with inverse f , i.e., $(f^{-1})^{-1} = f$;
- (3) if $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $g \in \text{Hom}_{\mathcal{C}}(B, C)$ are isomorphisms, then $g \circ f \in \text{Hom}_{\mathcal{C}}(A, C)$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof: ✓ ■

Definition 3. Let A, B be two objects in $\text{Obj}(\mathcal{C})$, if there is an isomorphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$, then we say A and B are **isomorphic** and denote $A \cong B$.

Example 8. Let $f \in \text{Hom}_{\text{Set}}(A, B)$ be an isomoprhim, then f is a bijection between the two sets A and B .

Definition 4. Let $\mathcal{C}$ be a category, a morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$ is called a **monomoprphism** if for any object Z in $\text{Obj}(\mathcal{C})$, we have

$$f \circ \alpha_1 = f \circ \alpha_2 \implies \alpha_1 = \alpha_2$$

for $\forall \alpha_1, \alpha_2 \in \text{Hom}_{\mathcal{C}}(Z, A)$.

Definition 5. Let $\mathcal{C}$ be a category, a morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$ is called a **epimoprphism** if for any object Z in $\text{Obj}(\mathcal{C})$, we have

$$\beta_1 \circ f = \beta_2 \circ f \implies \beta_1 = \beta_2$$

for $\forall \beta_1, \beta_2 \in \text{Hom}_{\mathcal{C}}(B, Z)$.

Example 9. In the category Set , the monomorphisms are precisely the injective maps between sets.

Proof: Let $f : A \longrightarrow B$ be an injective map between the two sets A and B . Suppose $\alpha_1, \alpha_2 : Z \longrightarrow A$ are two maps such that $f \circ \alpha_1 = f \circ \alpha_2 : Z \longrightarrow B$. Since f is injective, then f has a

left-inverse g such that $g \circ f = 1_A$. Hence we have

$$\alpha_2 = 1_A \circ \alpha_2 = (g \circ f) \circ \alpha_2 = g \circ (f \circ \alpha_2) = g \circ (f \circ \alpha_1) = (g \circ f) \circ \alpha_1 = 1_A \circ \alpha_1 = \alpha_1,$$

i.e., $f \in \text{Hom}_{\text{Set}}(A, B)$ is a monomorphism;

On the other hand, take $Z = \{z\}$ be a singleton and denote $a_1 := \alpha_1(z), a_2 := \alpha_2(z)$. Since $f \circ \alpha_1 = f \circ \alpha_2 \implies \alpha_1 = \alpha_2$, this means that

$$f(\alpha_1(z)) = f(\alpha_2(z)) \implies \alpha_1 = \alpha_2,$$

i.e.,

$$f(a_1) = f(a_2) \implies a_1 = a_2,$$

since $a_1 = \alpha_1(z) = \alpha_2(z) = a_2$. So f is injective. ■

Example 10. In the category Set , the epimorphisms are precisely the surjective maps between sets.

Proof: Let $f : A \longrightarrow B$ be surjective and suppose $\alpha_1 : B \longrightarrow Z, \alpha_2 : B \longrightarrow Z$ such that $\alpha_1 \circ f = \alpha_2 \circ f : A \longrightarrow Z$. Since f is surjective, so f has a right-side inverse g such that $f \circ g = 1_B$. So

$$\beta_1 = \beta_1 \circ 1_B = \beta_1 \circ (f \circ g) = (\beta_1 \circ f) \circ g = (\beta_2 \circ f) \circ g = \beta_2 \circ (f \circ g) = \beta_2 \circ 1_B = \beta_2;$$

On the other hand, take $Z = \{0, 1\}$ and define

$$\alpha_1 : B \longrightarrow \{0, 1\}$$

$$b \mapsto 0 \quad (\forall b \in B),$$

and

$$\alpha_2 : B \longrightarrow \{0, 1\}$$

$$b \mapsto \begin{cases} 0, & \text{if } b \in \text{Im} f, \\ 1, & \text{otherwise.} \end{cases}$$

then we see that $\alpha_1 \circ f(a) = \alpha_2 \circ f(a) = 0$ for all $a \in A$. Hence by condition, we get

$$\alpha_1 = \alpha_2 : B \longrightarrow \{0, 1\},$$

which implies that $B = \text{Im} f$, i.e., f is surjective. ■

3 Universal Properties

Definition 6. Let $\mathbf{C}$ be a category, an object I in $\text{Obj}(\mathbf{C})$ is called an **initial object** if for every object A in $\mathbf{C}$, there is exactly one morphism $I \longrightarrow A$ in $\mathbf{C}$, i.e.,

$$\text{Hom}_{\mathbf{C}}(I, A) \text{ is a singleton for every object } A \text{ in } \mathbf{C}.$$

Definition 7. Let $\mathbf{C}$ be a category, an object F in $\text{Obj}(\mathbf{C})$ is called a **final object** if for every object A in $\mathbf{C}$, there is exactly one morphism $A \longrightarrow F$ in $\mathbf{C}$, i.e.,

$$\text{Hom}_{\mathbf{C}}(A, F) \text{ is a singleton for every object } A \text{ in } \mathbf{C}.$$

Remark: A category need not have initial objects or final objects. Initial and final objects may not be unique when they exist.

Example 11. In the category **Set**, the empty set $\emptyset$ is the initial object and every singleton $\{p\}$ is a final object.

Although the initial and final objects may not be unique when they exist, however, they are unique up to isomorphism.

Proposition 3. Let $\mathbf{C}$ be a category,

- (1) if I_1 and I_2 are both initial objects in $\mathbf{C}$, then $I_1 \cong I_2$;
- (2) if F_1 and F_2 are both final objects in $\mathbf{C}$, then $F_1 \cong F_2$.

Proof: (1) Since I_1 is initial, then we see that $\text{Hom}_{\mathbf{C}}(I_1, I_1)$ consists of only 1_{I_1} , the identity morphism. Suppose $f \in \text{Hom}_{\mathbf{C}}(I_1, I_2)$ and $g \in \text{Hom}_{\mathbf{C}}(I_2, I_1)$ are the only morphisms for the initial objects I_1 and I_2 , then we get the composition morphism

$$f \circ g \in \text{Hom}_{\mathbf{C}}(I_2, I_2) \text{ and } g \circ f \in \text{Hom}_{\mathbf{C}}(I_1, I_1),$$

which must be

$$f \circ g = 1_{I_2} \text{ and } g \circ f = 1_{I_1},$$

which means that $I_1 \cong I_2$.

(2) similar to (1). ■